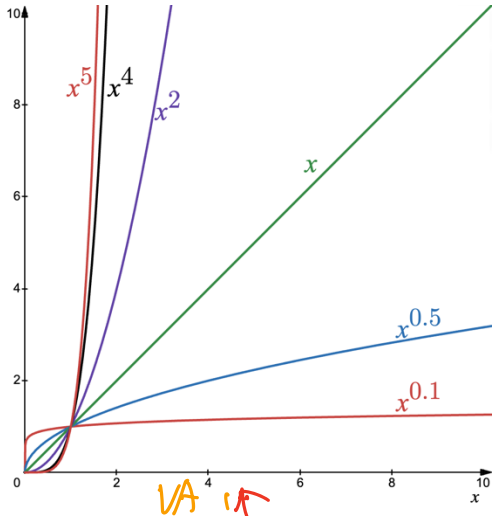
MAT1375, Classwork11, Fall2026

Ch10. Rational Functions II



6. The order of growth for functions: **Power functions**.

$$x^p, p > 0$$

For each $x > 1$, we have

$$\cdots < x^{0.1} < x^{0.5} < x < x^2 < x^4 < x^5 < \cdots$$

7. The definition of a **Horizontal Asymptote**:

The line $\boldsymbol{y = b}$ is a horizontal asymptote or the graph of a function f if $f(x)$ approaches $\boldsymbol{b}$ as $\boldsymbol{x}$ **increases** or **decreases without bound**.

8. What is the difference of Vertical Asymptote and Horizontal Asymptote?

Vertical Asymptote $x = c$ when $x \to c$ $f(x) \to \infty$ or $f(x) \to -\infty$

VA $x = c$

Horizontal Asymptote $y = \frac{P}{q}$ when $x \to \infty$ or $x \to -\infty$, $f(x) \to \frac{P}{q}$

9. How to locate Horizontal Asymptotes: Let $f(x) = \frac{p(x)}{q(x)}$ be a rational function given by

$$f(x) = \frac{p_n x^n + p_{n-1}x^{n-1} + \cdots + p_1 x + p_0}{q_m x^m + q_{m-1}x^{m-1} + \cdots + q_1 x + q_0}, p_n \neq 0, q_m \neq 0.$$

deg n, deg m

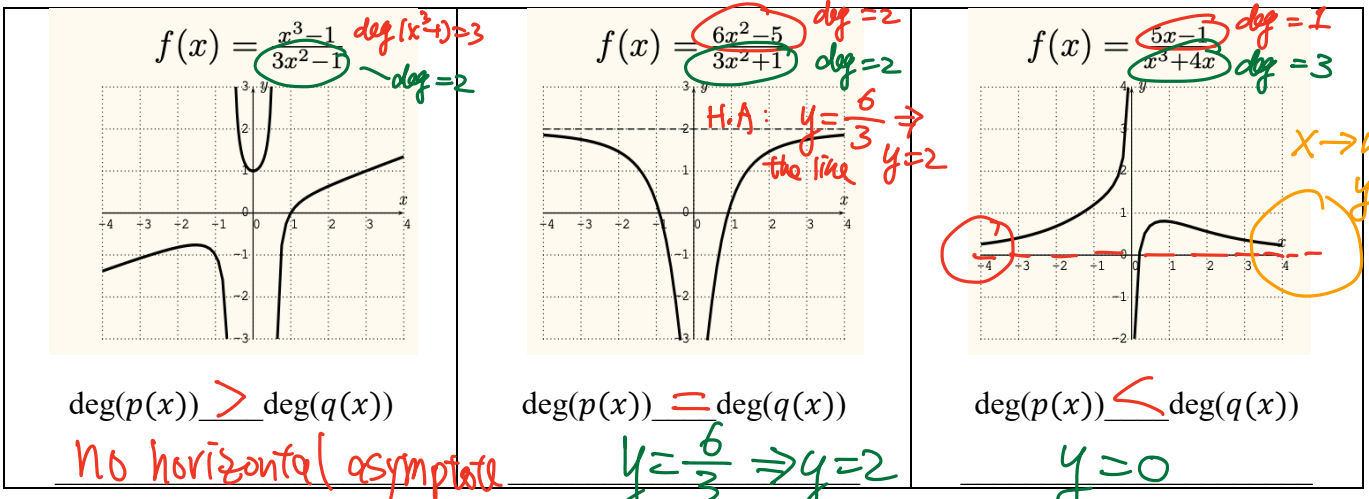
where the degree of the numerator is n and the degree of the denominator is m.

(1) If $n > m$, the graph of f has no horizontal asymptote.

(2) If $n = m$, the line $y = \frac{p_n}{q_m}$ (which is the ration of two leading coefficients) is the horizontal asymptote of the graph of f.

~~$x = 0$~~

(3) If $n < m$, the x-axis (which is $y = 0$) is the horizontal asymptote of the graph of f.



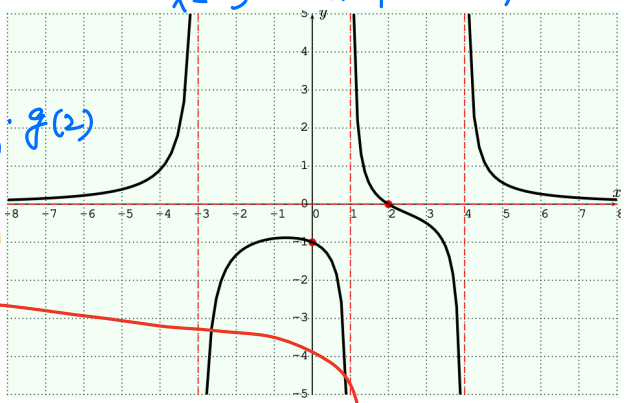
$\deg(p(x))$ > $\deg(q(x))$ — no horizontal asymptote

$\deg(p(x))$ = $\deg(q(x))$ — $y = \frac{6}{3} \Rightarrow y = 2$

$\deg(p(x))$ < $\deg(q(x))$ — $y = 0$

Ch11. More on Rational Functions

1. The graph of $f(x) = \frac{p(x)}{g(x)}$ is displayed below, where $\deg(p(x)) = 1$ and $\deg(g(x)) = 3$.



Find the intercepts, asymptotes, and a formula for $f(x)$.

For 2. And 3., let $f(x) = \frac{p(x)}{g(x)}$ be a rational function and **$\deg(p(x)) > \deg(g(x))$**.

2. Rational Function and Long Division:

If $p(x)$ divided by $g(x)$ can be represented with a quotient $q(x)$ and a remainder $r(x)$ where $\deg(r(x)) <$ ____ $\deg(g(x))$, one can rewrite $f(x)$ as

$$f(x) = \frac{p(x)}{g(x)} = \text{____} + \frac{r(x)}{g(x)}.$$

3. Asymptotic Behavior with Slant Asymptote:

Since $\deg(r(x)) < \deg(g(x))$, for large $|x|$ (which is $x \to \pm$ ____), we have

$\frac{r(x)}{g(x)}$ approaches _____ so that $f(x)$_____$q(x)$.

If $q(x)$ is a linear function (which is a polynomial of degree _____), then q is called the _____ asymptote of f.

4. Find the **slant asymptote** of the rational function $f(x) = \frac{2x^3 - 13x^2 + 35x - 26}{x^2 - 4x + 6}$.